

RESCUE JUNIOR - Bill of Materials (BOM)

Team name: Tuton

Every Line/Maze team has to submit a bill of materials for their robot

Instructions:

*Enter Team name above
*All costs need to be in local currency and their approximate conversion into dollars.
*All components worth less can be summarized in one line quantity 1 and their total cost (e.g. Screws)
*In the Local Currency Name column, the team must enter the name of the local currency, if it is different from US dollars.
*In the software sheet, in the Author column, the name of the designer, writer, or company that owns the software or library should be entered. If the team created their own algorithm, dataset, or AI model, they can include their names as authors.
*On the hardware sheet, in the "Author" column, enter the name of the company, engineer, or part manufacturer. If the team made a custom part, such as the robot housing, the team should include their names as the authors.

IMPORTANT The "Hardware" sheet is not required for simulation

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Name of the Local Currency	Total Cost Local Currency	Total Cost (\$ U.S.A. Dollars
Japanese Yen JPY	¥126,514	\$784.39

#	Component	Part name	Author	Source	Quantity	Unit cost (Local Currency)	Unit cost (US Dollars)	(Local	(USA Dollars)
1	PCB with electronic components	Custom made	Shuji Igarashi / JLCPCB	https://jlcpcb.com/	1	¥6,000	\$37.20	¥6,000	\$37.20
2	Metall parts for frame	Custom made	Ryouta Koudu / JLCPCB	https://jlcpcb.com/	1	¥5,000	\$31.00	¥5,000	\$31.00
3	Plastic parts for frame	Custom made	Ryouta Koudu	3D Printer	1	¥600	\$3.72	¥600	\$3.72
4	Connecting parts	Screws, nuts, washers, ...		Local tool store	1	¥1,200	\$7.44	¥1,200	\$7.44
5	Cables	Various cables	Ryouta Koudu	https://www.jst-mfg.com/	1	¥2,000	\$12.40	¥2,000	\$12.40
6	Wheels	Custom made	Ryouta Koudu	http://www.engravingjapan.com/	4	¥100	\$0.62	¥400	\$2.48
7	Drive motors	STS3032	Feetech RC MODEL CO., Ltd.	https://www.feetechrc.com/	4	¥4,800	\$29.76	¥19,200	\$119.04
8	Motors for rescue kits	MG90D	Tower Pro Pte Ltd	https://towerpro.com.tw/	2	¥1,350	\$8.37	¥2,700	\$16.74
9	LED	WS2812B	Worldsemi Co.,Ltd	http://www.world-semi.com/	31	¥25	\$0.16	¥775	\$4.81
10	LCD Display	M154-240240-RGB	SUNHOKEY Electronics Co.,Ltd.	https://sunhokey.cn/	1	¥750	\$4.65	¥750	\$4.65
11	Buzzer	UGCT7525AN4	UNIVERSAL(CHANGZHOU) ELECTRONICS CO.,LTD	https://www.china-speakers-manufacturer.com/	1	¥280	\$1.74	¥280	\$1.74
12	Color sensor	S9706	Hamamatsu Photonics K.K.	https://www.hamamatsu.com/jp/en.html	1	¥600	\$3.72	¥600	\$3.72
13	Reflectance sensor	LBR127HLD	Letex Technology Corp.	https://www.letex.com.tw/index.aspx	2	¥80	\$0.50	¥160	\$0.99
14	Switch	Various switches	Cosland Co., Ltd.	https://www.cosland.com/about.php	1	¥500	\$3.10	¥500	\$3.10
15	9-axis IMU Sensor	BNO085	Adafruit Industries	https://www.adafruit.com/	1	¥4,074	\$25.26	¥4,074	\$25.26
16	Camera for victims	UnitV M12 Camera	M5Stack Technology Co., Ltd.	https://m5stack.com/	2	¥7,282	\$45.15	¥14,564	\$90.30
17	Camera for color tiles	Raspberry Pi Camera V3	Raspberry Pi Foundation	https://www.raspberrypi.org/	1	¥6,000	\$37.20	¥6,000	\$37.20
18	ToF Distance Sensor	VL53L4CX	STMicroelectronics	https://www.st.com/content/st_com/en.html	4	¥2,000	\$12.40	¥8,000	\$49.60
19	Single point LiDAR	TSD10	PONO Technology Co.,Ltd.	https://positive-inno.com/	4	¥1,980	\$12.28	¥7,920	\$49.10
20	360° LiDAR	T-mini Plus	Shenzhen EAI Technology Co.,Ltd.	https://www.ydlidar.com/	1	¥14,800	\$91.76	¥14,800	\$91.76
21	Single Board Computer	Raspberry Pi 4 model B	Raspberry Pi Foundation	https://www.raspberrypi.org/	1	¥23,210	\$143.90	¥23,210	\$143.90
22	STM32 Microcontroller	STM32F446RET6	STMicroelectronics	https://www.st.com/content/st_com/en.html	3	¥1,467	\$9.10	¥4,401	\$27.29
23	STM32 Microcontroller	STM32F103C8T6	STMicroelectronics	https://www.st.com/content/st_com/en.html	1	¥1,000	\$6.20	¥1,000	\$6.20
24	Battery	LiPo 11V 1300mAh	Kypom Technology Stock Co., Ltd	http://kypom-battery.com/	1	¥2,380	\$14.76	¥2,380	\$14.76

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Name of the Local Currency	Total Cost Local Currency	Total Cost (\$) U.S.A. Dollars
Japanese Yen JPY	¥0	\$0.00

#	Name	Software's Tool/Library	Description	Source	Autor	Cost (Local Currency)	Cost (USA Dollars)
1	Visual Studio Code	Software's Tool	Programmig (Raspberry Pi code)	https://code.visualstudio.com/	Microsoft	FREE	FREE
2	Git, GitHub	Software's Tool	Version control for managing functions	https://github.co.jp/	GitHub, Inc.	FREE	FREE
3	MaixPy IDE	Software's Tool	Programmig (STM32)	https://wiki.sipeed.com/maixpy/en/	Sipeed	FREE	FREE
4	Autodesk Fusion	Software's Tool	3D CAD shell design	https://www.autodesk.com/jp/products/fusion-360/overview	Autodesk Inc.	FREE (Education)	FREE (Education)
5	KiCAD 9.0	Software's Tool	PCB design	https://www.kicad.org/	OSS Contributors	FREE	FREE
6	Adafruit NeoPixel	Library	For controlling WS2812B LEDs	https://github.com/adafruit/Adafruit_NeoPixel	Adafruit Industries	FREE	FREE
7	Adafruit BNO08x	Library	For controlling BNO085 IMU Sensor	https://github.com/adafruit/Adafruit_BNO08x	Adafruit Industries	FREE	FREE
8	Adafruit SSD1306	Library	For controlling OLED display	https://github.com/adafruit/Adafruit_SSD1306	Adafruit Industries	FREE	FREE
9	STM32duino VL53L4CX	Library	For controlling VL53L4CX distance sensors	https://github.com/stm32duino/VL53L4CX	OSS Contributors	FREE	FREE
10	SCServo	Library	For controlling STS3032 Servos	https://github.com/workloads/scservo	Kerim Satirli	FREE	FREE
11	OpenCV	Library	Image Processing	https://pypi.org/project/opencv-python/	Foundation/OSS	FREE	FREE
12	picamera2	Library	Image Processing	https://github.com/raspberrypi/picamera2	Raspberry Pi Ltd	FREE	FREE
13	Numpy	Library	Array and math library	https://pypi.org/project/numpy/	OSS Contributors	FREE	FREE
14	ydliar	Library	For controlling T-mini Plus LiDAR sensor	https://github.com/YDLIDAR/YDLidar-SDK	Technology Co.,Ltd.	FREE	FREE
15	Matplotlib	Library	graphics generation	https://matplotlib.org/	OSS Contributors	FREE	FREE